Sustainable Development Goals (SDGs) and Climate Change

WI1103 Introduction to the Principles of Sustainability

Week 6



6.1. Climate Change and the Achievement of Sustainable Development Goals

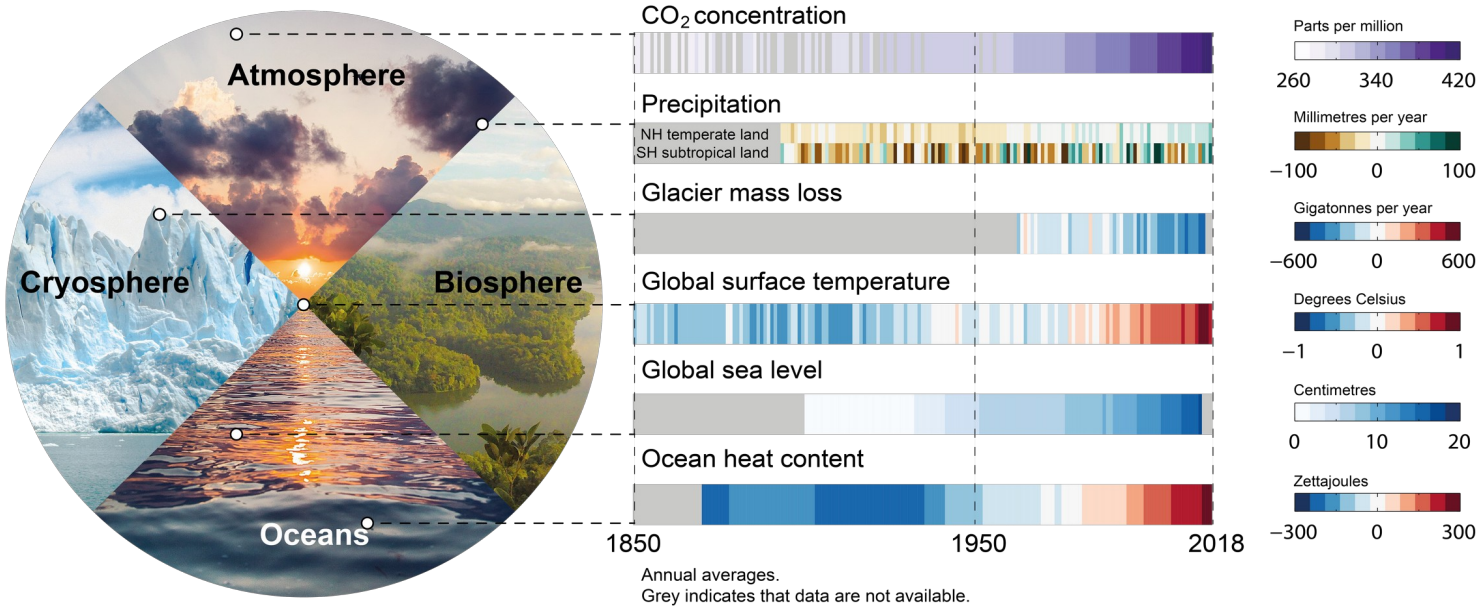
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What is Climate Change?

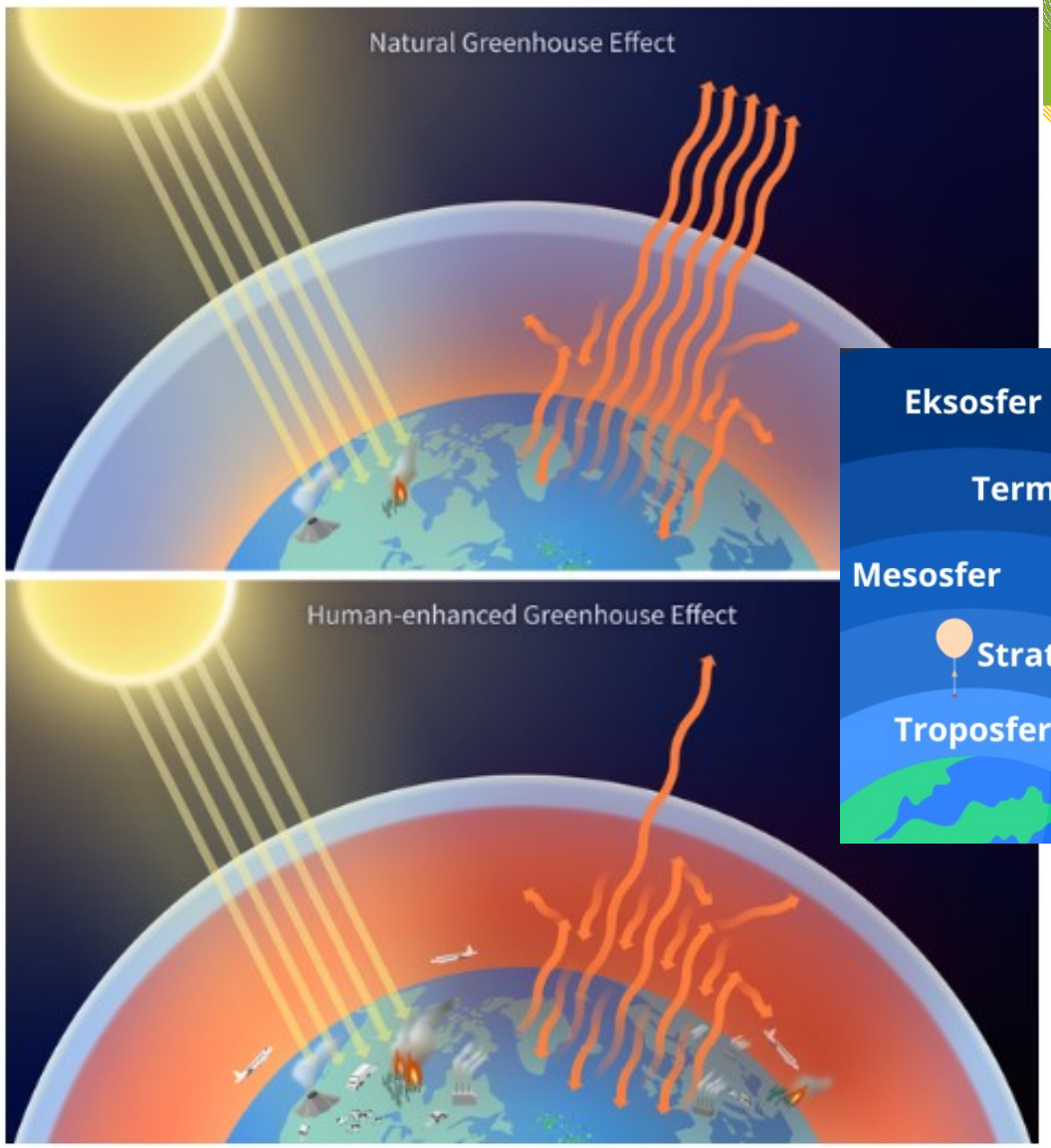
- Climate is the average atmospheric condition in a region over a long period of time.
- Climate change is the long-term change in the average atmospheric conditions that shape the local, regional, and global climate of the Earth.
- Throughout Earth's history, the climate has changed due to natural factors such as volcanic activity, variations in solar radiation, tectonic plate shifts, and orbital changes.
- The climate change occurring today is happening at an extraordinary speed and cannot be explained solely by natural factors, but is primarily driven by human activities.



Changes are occurring throughout the climate system. The main components of the climate system are the atmosphere, biosphere, cryosphere, and oceans (IPCC, 2021)

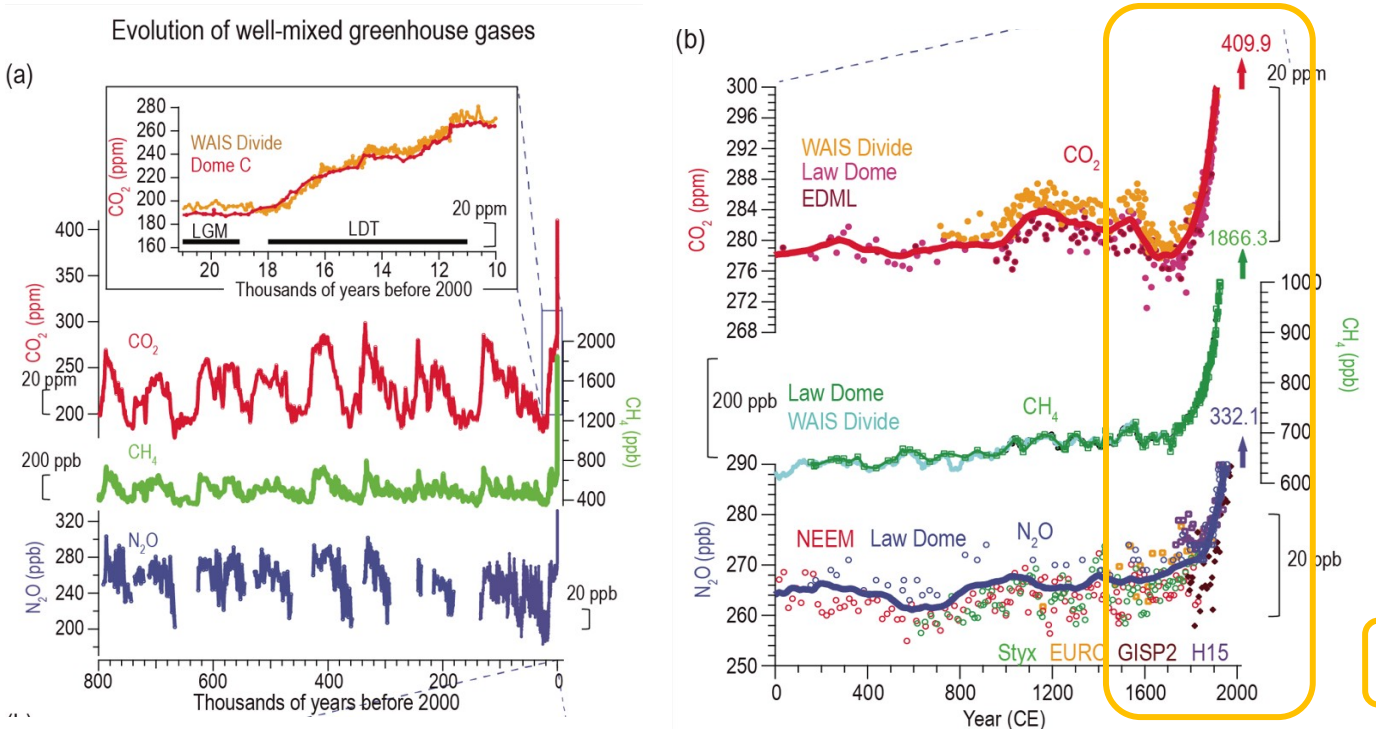
Greenhouse Gases (GHGs)

- Greenhouse gases (GHGs) are a group of gases that absorb heat energy emitted from the Earth's surface.
- Greenhouse gases in ideal amounts are important because the natural greenhouse effect they produce keeps the Earth's average surface temperature above freezing, thus allowing life to exist.
- There are four important greenhouse gases present on Earth:
  - - Water vapor
  - - Carbon dioxide
  - - Methane
  - - Nitrous oxide
  - - Ozone
- Other GHGs caused by human activities include gases from industries such as CFCs, HFCs, PFCs, and SF6.



Greenhouse effect diagram. Credit: NOAA Climate Program Office, Graphic by Anna Eshelman

Changes in GHGs concentration over time



- Recorded greenhouse gases (GHGs) concentrations over the last 800,000 years (left image)
- A sharp increase in GHGs concentrations occurred after the Industrial Revolution (~1760).
- CO_2 in 2019 was recorded at 409.9 ppm (right chart)

- Concentration of 3 GHGs (Carbon Dioxide, Methane, and Nitrous Oxide) in ppm except Methane in ppb.

Reference period (*See Interactive Atlas for climate model output)	Age	CO_2 (ppm)	Temperature (°C)	Sea level (m)
Recent past	1995–2014 CE	360 ⟶ 397	0.66 to 1.00	0.15 to 0.25
Approximate pre-industrial	1850–1900 CE	286 ⟶ 296	−0.15 to +0.11	−0.03 to 0.00
Last Millennium	850–1850 CE	278 to 285	−0.14 ~ 0.24	−0.05 to 0.03
Mid-Holocene*	6.5–5.5 ka	260 to 268	0.2 to 1.0	−3.5 to +0.5
Last Deglacial Transition	18–11 ka	193 ⟶ 271	not assessed	−120 ⟶ −50
Last Glacial Maximum*	23–19 ka	188 to 194	−5 to −7	−134 to −125
Last Interglacial*	129–116 ka	266 to 282	0.5 to 1.5	5 to 10
Mid-Pliocene Warm Period*	3.3–3.0 Ma	360 to 420	2.5 to 4.0	5 to 25
Early Eocene	53–49 Ma	1150 to 2500	10 to 18	70 to 76
Paleocene-Eocene Thermal Maximum	55.9–55.7 Ma	900 ⟶ 2000	10 to 25	not assessed

X to Y: *very likely* range (caveats in Figure 2.34)
X ⟶ Y: start to end of period, with no stated uncertainty
X ~ Y: lowest and highest values, with not stated uncertainty

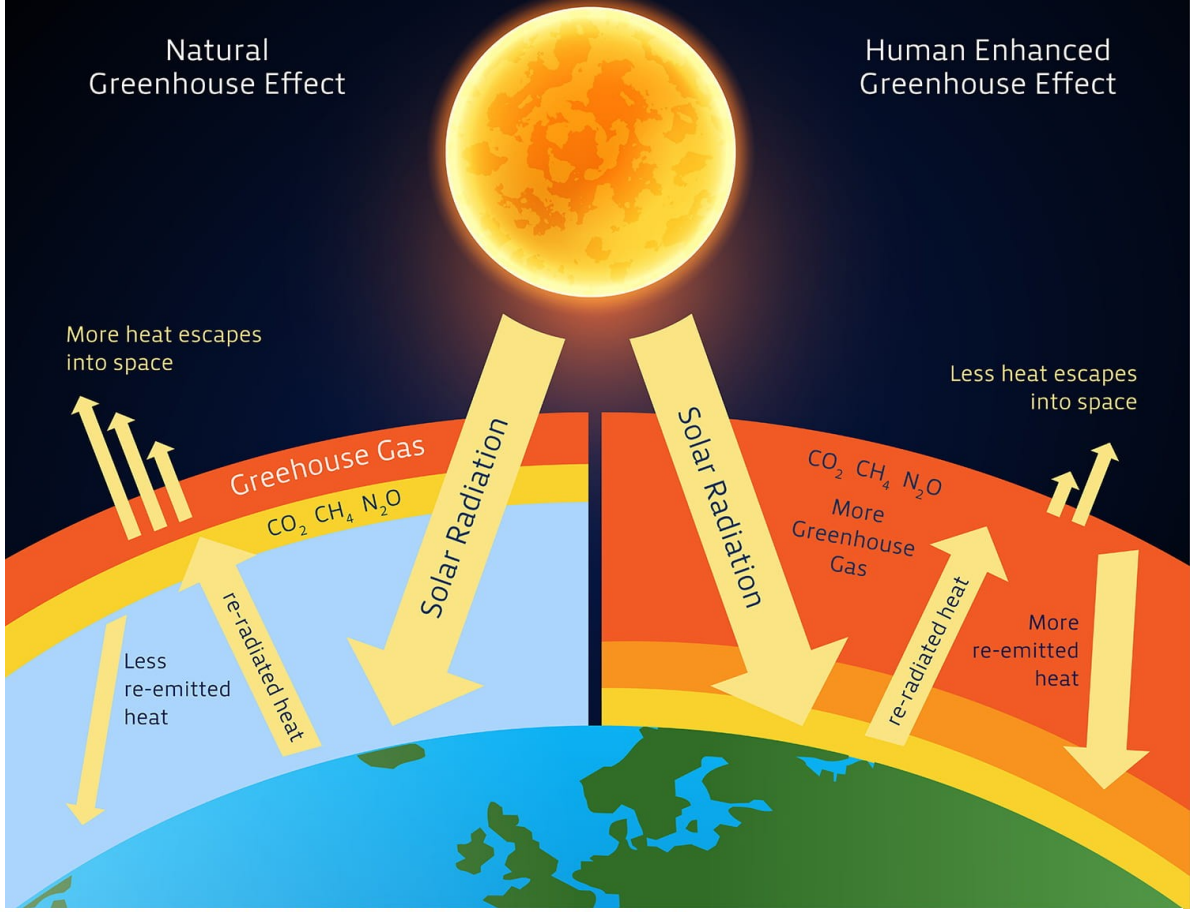
lower 1850–1900 higher

Excessive GHGs emissions - Global Warming

With the increase of greenhouse gases, such as carbon dioxide and methane, produced from human activities into the atmosphere, it causes the average global temperature to rise.

Enhanced greenhouse gas effect--> global warming

GHGs have different GWP (Global Warming Potential).



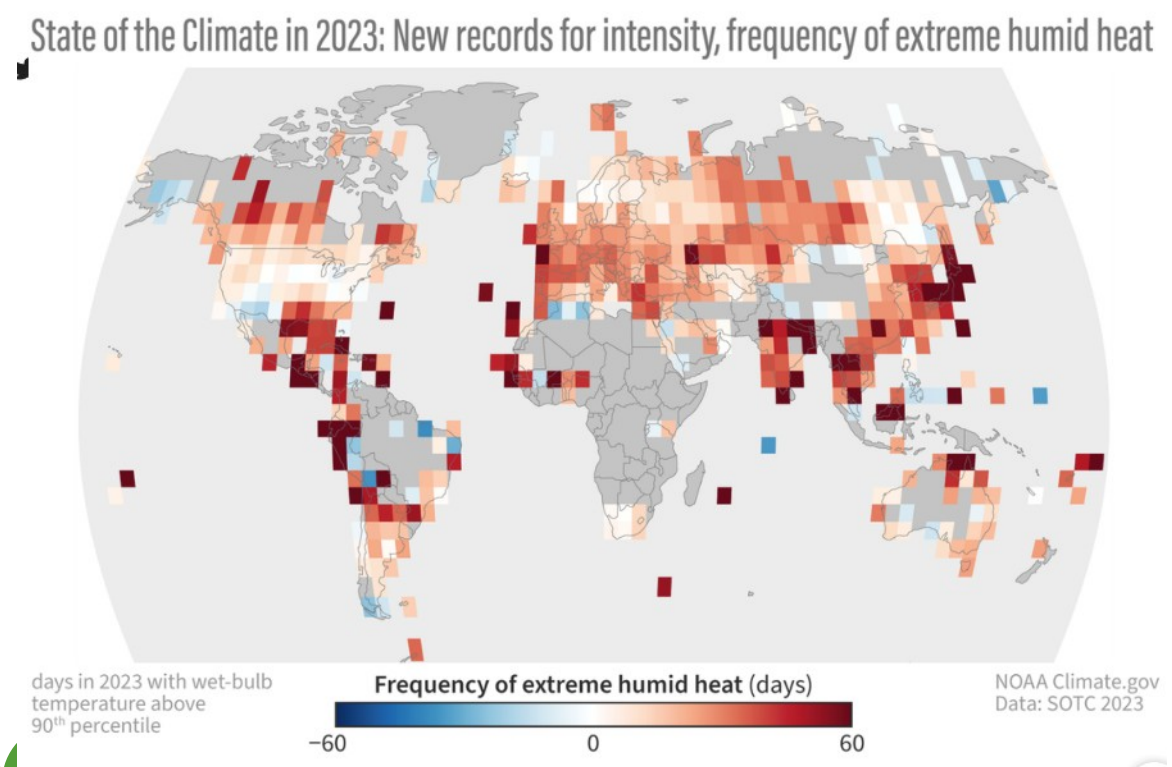
IPCC Global Warming Potential (GWP) values relative to CO_2

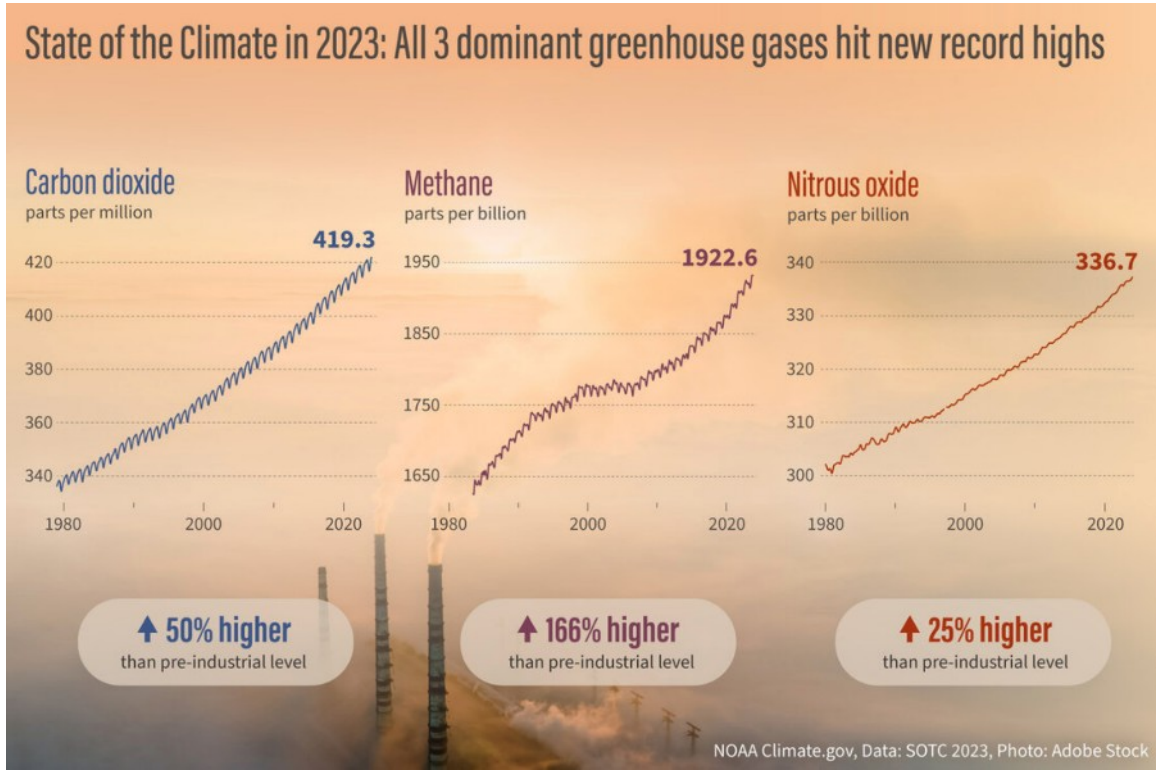
Common chemical name or industrial designation	Chemical formula	GWP values for 100-year time horizon		
		Fourth Assessment Report (AR4)	Fifth Assessment Report (AR5)	Sixth Assessment Report (AR6)
Major Greenhouse Gases				
Carbon dioxide	CO_2	1	1	1
Methane – non-fossil	CH_4	25	28	27.0
Methane – fossil	CH_4	N/A	30	29.8
Nitrous oxide	N_2O	298	265	273
Nitrogen trifluoride	NF_3	17,200	16,100	17,400
Sulfur hexafluoride	SF_6	22,800	23,500	24,300

IPCC Global Warming Potential (GWP) values relative to CO_2

Common chemical name or industrial designation	Chemical formula	GWP values for 100-year time horizon		
		Fourth Assessment Report (AR4)	Fifth Assessment Report (AR5)	Sixth Assessment Report (AR6)
Hydrofluorocarbons (includes unsaturated hydrofluorocarbons)*				
HFC-23	CHF_3	14,800	12,400	14,600
HFC-32	CH_2F_2	675	677	771
HFC-41	CH_3F	92	116	135
HFC-125	CHF_2CF_3	3,500	3,170	3,740
HFC-134	CHF_2CHF_2	1,100	1,120	1,260

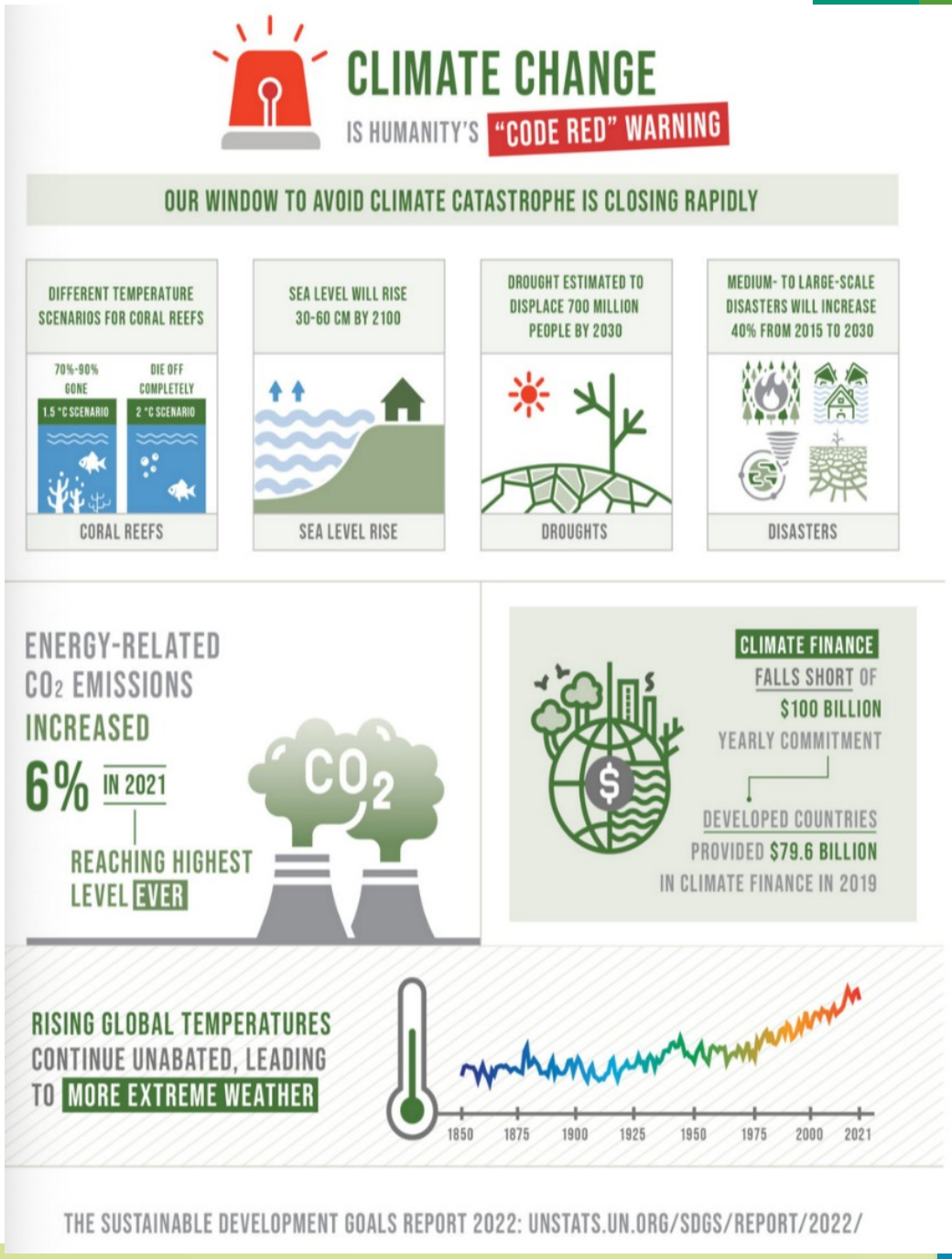
Related Data





Challenges in Achieving Sustainable Development Goals (SDGs)

- What are the causes of the difficulty in achieving the SDGs?
- According to the report by the Intergovernmental Oceanographic Commission (IOC) & World Meteorological Organization (WMO) ~ Climate change has the potential to hinder the achievement of other SDGs (reserved progress).
- The impacts of weather, climate, and water cause disasters (slow and fast onset):
  - Temperature increase
  - Sea level rise
  - Disruption of ecosystems (land and sea)



From SDG 13 other SDGs

- Extreme weather events ~ as a cause
- COVID-19 pandemic ~ reduced GHG emissions by 6%
- Vulnerable groups:
  - Small Island Developing States (SIDS)
  - Coastal areas
  - Countries in the Global South
  - Vulnerable groups, women, children, youth, and other marginalized groups



Other SDG contributions to SDG 13

- Climate change affects the achievement of other SDGs.
  There needs to be synergy between the achievement of SDG 13 and other SDGs.
- On the other hand, the achievement of other SDGs has the potential to help achieve SDG 13.

For example:

- When human activities produce greenhouse gases, about half of these emissions remain in the atmosphere, while the other half is absorbed by land and oceans.

- Ecosystems and the biodiversity within them function as natural carbon sinks.

Biodiversity - our strongest natural defense against climate change



TARGET 15.3

END DESERTIFICATION AND RESTORE DEGRADED LAND

By 2030, combat desertification, restore degraded land and soil, including land affected by desertification, drought and floods, and strive to achieve a land degradation-neutral world.

TARGET 15.5

PROTECT BIODIVERSITY AND NATURAL HABITATS

Take urgent and significant action to reduce the degradation of natural habitats, halt the loss of biodiversity and, by 2020, protect and prevent the extinction of threatened species.

TARGET 14.2

PROTECT AND RESTORE ECOSYSTEMS

By 2020, sustainably manage and protect marine and coastal ecosystems to avoid significant adverse impacts, including by strengthening their resilience, and take action for their restoration in order to achieve healthy and productive oceans.

2025: Mid Critical Time

- Each country reports its emission reduction contributions through Nationally Determined Contributions (NDCs)
- Increasing hunger and declining food security
- Slow progress in climate adaptation
- Synergy between climate change and development financing



Follow Up

- There is a need for the development of science related to climate, weather, and water resources.
- Stocktaking on Climate Change.





6.2 The 2015-2030 Paris Agreement and Indonesia's Commitment

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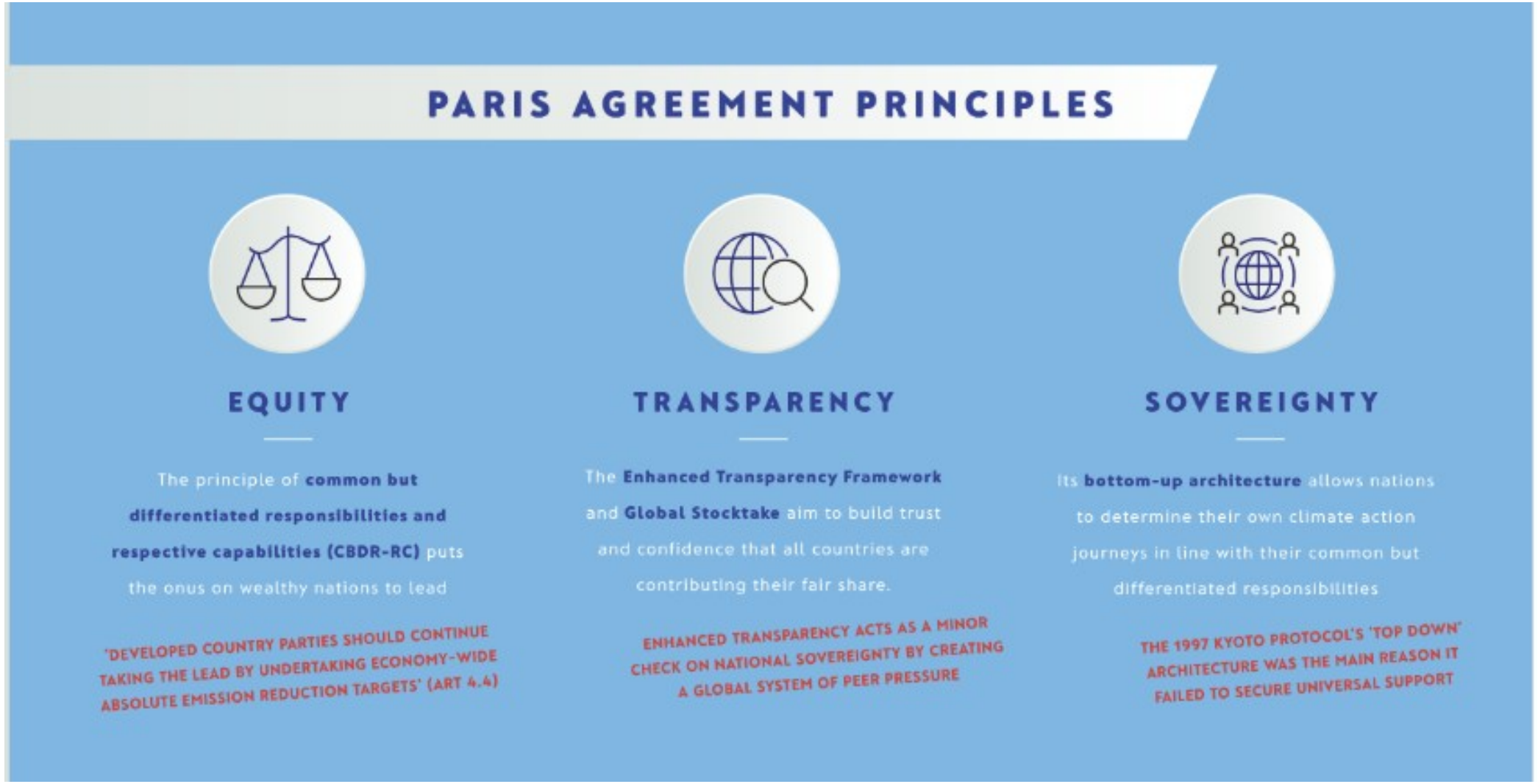
Paris Agreement 2015-2030







Paris Agreement Principles



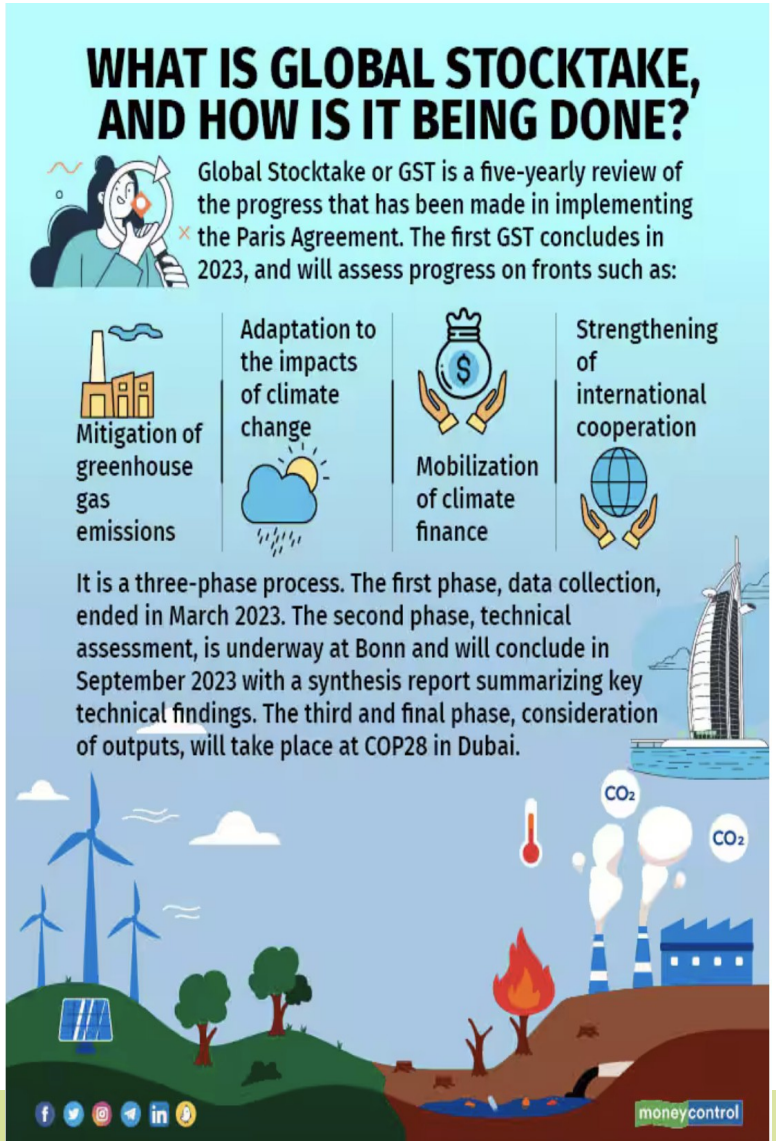
Prinsipal Equity: ”*Common but Differentiated Responsibilities & Respective Capabilities(CBDR-RC)*”

- Negotiation terminology: ‘Right to Development’
- Common responsibilities
  - All countries contribute to Climate Change (to varying degrees)
  - Common responsibility to save the Earth
- Differentiated Responsibilities
  - Principle of ‘ability to pay’ ~ Developed countries commit more
  - Considering the conditions & Development goals of each country
- Respective Capabilities
  - Differences in each country's capacity to face environmental challenges –(Conference Statement: TFM, Funding to enable global participation)



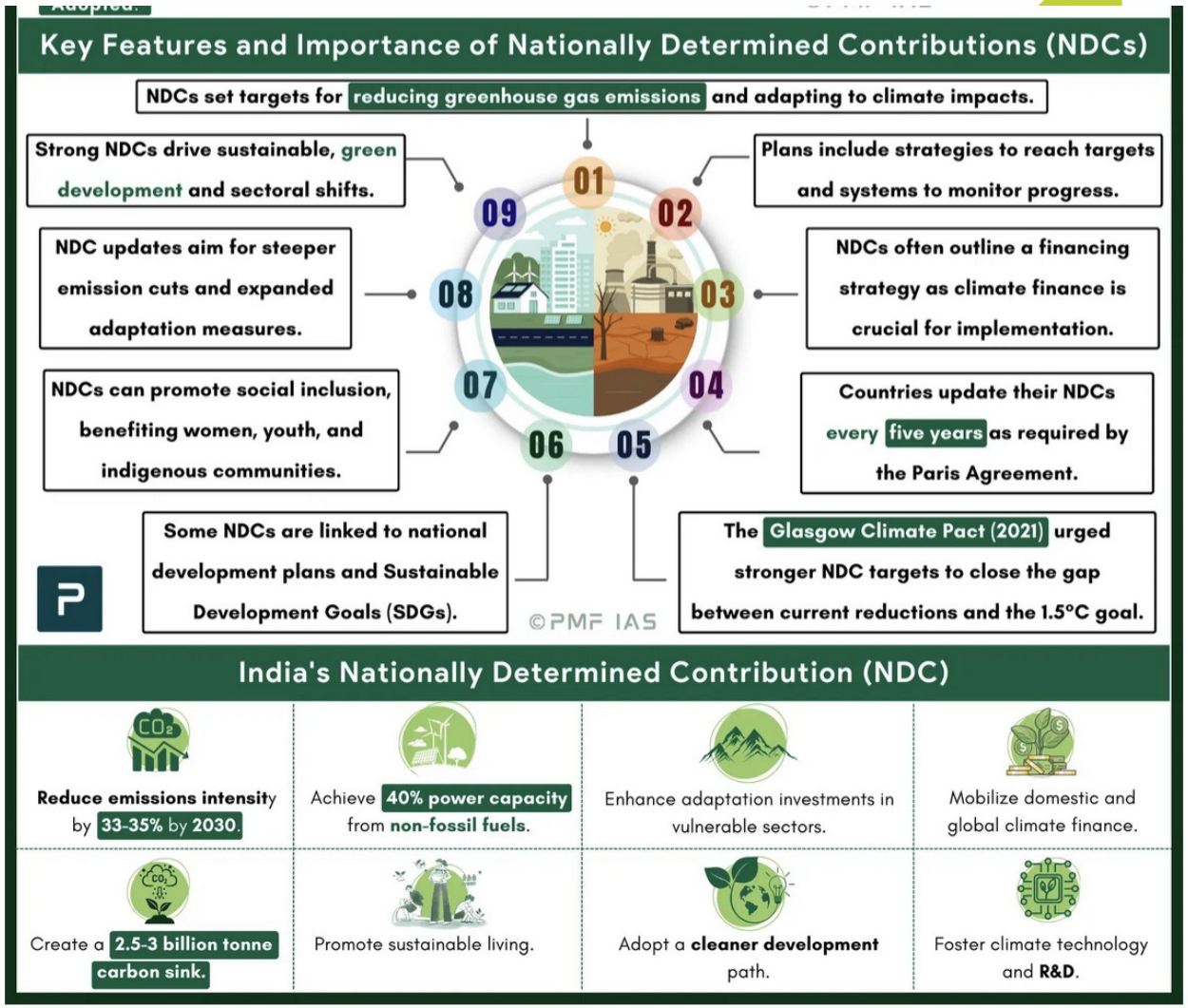
Principle of Transparency: Global Stocktake (GST)

- Assess the extent to which various countries have implemented their climate change commitments (review progress).
- Objective: to continually increase climate ambition to avoid an emergency situation.
- Examples: reduce fossil fuels, increase renewable energy capacity, increase climate funding.



Principle of Sovereignty

- Bottom-up Architecture
- Allows each country to determine its own level of emissions
- Reported as Nationally Determined Contributions (NDC)



Indonesia's Commitment

- The Indonesian government ratified the Paris Agreement as Law No. 16/2016.
  - The 2030 GHG emission reduction target is 29% independently and 41% with international support.
  - Limiting the increase in Earth's temperature to < 1.5℃.
  - Achieving Net Zero Emission (NZE) by 2060 or earlier.



2nd NDC – Indonesia 2025

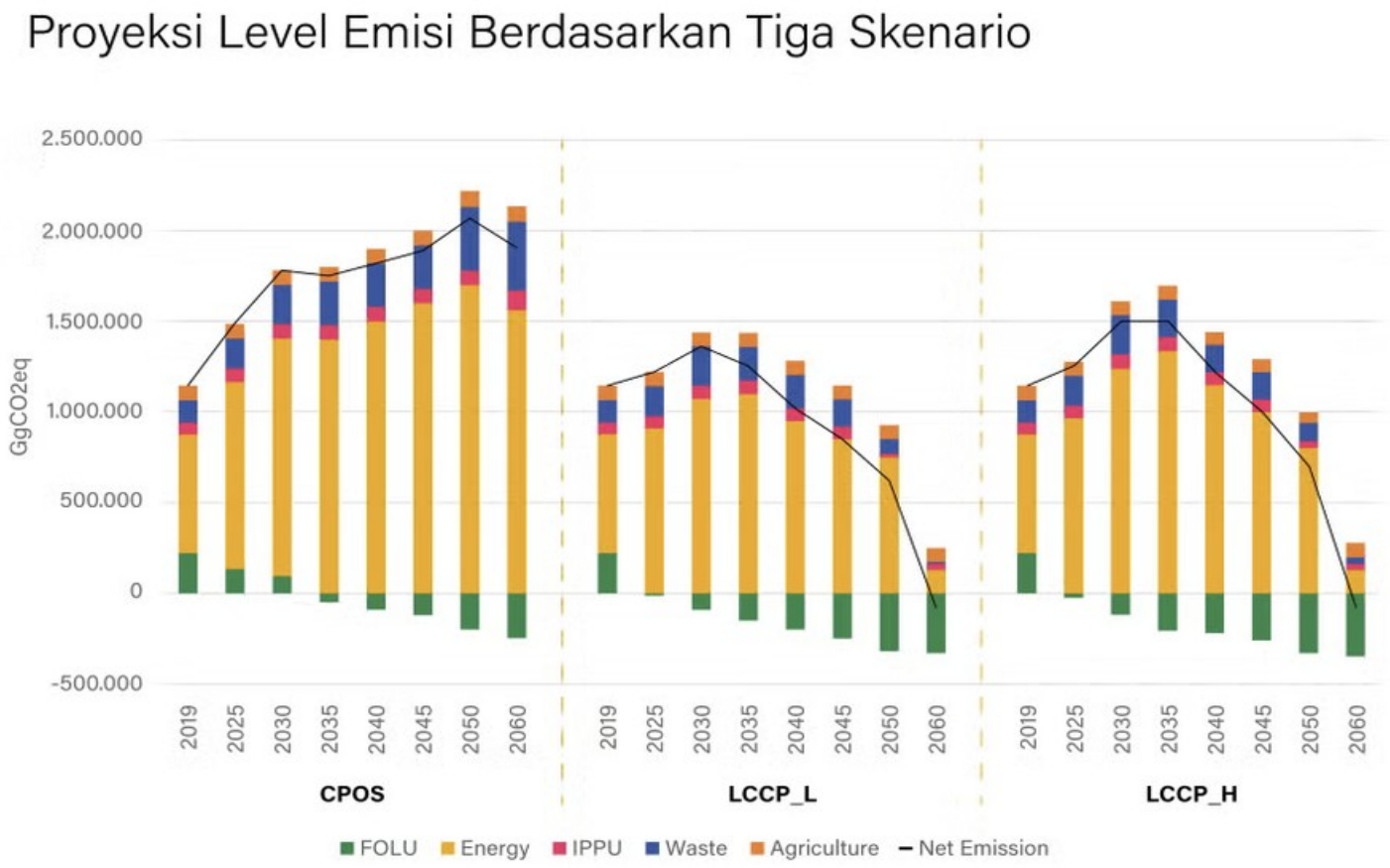
- **In response to the outcomes of the 1st Global Stocktake:**
- The absolute emission reduction targets for 2035 are **31.89%** (unconditional / without intervention) and **43.20%** (conditional / with intervention).
- **New Scenarios:**
- **CPOS:** Current Policy Scenario
- **LCCP_Low:** Low Carbon Compatible Paris Agreement (Low)
- **LCCP_High:** Low Carbon Compatible Paris Agreement (High)

ENDC	Perubahan Utama	SNDC
2020-2030	Perpanjangan Periode	2031-2035
Persentase pengurangan emisi pada tahun 2030: 31,89% (tanpa syarat) dan 43,2% (bersyarat) di bawah BAU.	Pembaruan Kejelasan	**Emisi absolut** tahun 2035 di bawah skenario LCCP: ▪ **1,257,717 Gg CO_2e (LCCP_L)** ▪ **1,488,866 Gg CO_2e (LCCP_H)**
▪ Pendekatan ***Business as Usual*** dengan 2010 sebagai tahun acuan ▪ **GWP IPCC AR 2**	Pembaruan Tahun Dasar dan Metodologi	▪ **2019 sebagai tahun acuan** ▪ **GWP IPCC AR 5** (sesuai dengan UNFCCC *transparency standards* terbaru) ▪ **Sesuai dengan Keputusan Dec.1/CMA 5** (Keluaran global)
CO_2, CH_4, N_2O	Perluasan Lingkup GRK	**CO_2, CH_4, N_2O, HFC**
Sekitar **USD 247 miliar** dari tahun 2018 hingga 2030 menurut Laporan BUR Kedua.	Dukungan Kebutuhan	Perkiraan awal sebesar **USD 472,6 miliar** antara tahun 2030–2035 di sektor energi, pertanian, FOLU, dan limbah (tidak termasuk IPPU)
FOLU, pertanian, energi, IPPU, dan limbah	Perluasan Lingkup	**Lima sektor awal dengan subsektor tambahan:** sektor hulu minyak dan gas di bawah sektor energi, gas fluorinated (*F-gases*) di bawah sektor IPPU, potensi ekosistem pesisir dan karbon biru.
▪ **BaU:** *Business as Usual* ▪ **CM1 (Skenario Tanpa Syarat)** dengan upaya domestik saja ▪ **CM2 (skenario bersyarat)** dengan bantuan internasional	Pembaruan Trajektori	▪ **CPOS:** Kebijakan saat ini, tanpa dukungan internasional (pertumbuhan moderat: 6,0% pada 2030; 6,7% pada 2035). ▪ **LCCP_L:** Selaras dengan target LTS-LCCR dan Perjanjian Paris (pertumbuhan moderat: 6,0% pada 2030; 6,7% pada 2035). ▪ **LCCP_H:** Selaras dengan target LTS-LCCR dan Perjanjian Paris (pertumbuhan tinggi: 7,0% pada 2030; 8,3% pada 2035)

WRI INDONESIA

Emision Pjojection

- Scenarios 2 & 3 show emissions peaking in 2030, before dropping to become a net sink by 2060.
- Because it is conditional, it depends on international funding.



WRI INDONESIA

6.3 Examples

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Kiribati, The Country That Will Sink?

- With a population of 100,000 people, 33 atolls, islands in Polynesia
- Flooding since the 1970s
  President Anote Tong (2015): “Migrate with Dignity”: Purchased 6,000 acres in Fiji
- President Taneti Maamau (2022): ‘fortifying the islands’ by building a seawall ~ reason: climate crisis is happening everywhere





An aerial view of one of Kiribati’s islands. © Matthieu Rytz



Semarang: Prevent Abrasion

- Infrastructure investment:
  Coastal dike
  Sea dike
  Maritime toll road
  Mangrove forest
  Dam from tires









Muara Gembong, Kabupaten Bekasi

- Desa Pantai Mekar
- Those who remain: Declining quality of life
- Forced to relocate (displacement): Starting over





The Phenomenon of Rising Heat in Urban Areas

- Heat waves are occurring in Europe, Southeast Asia, and other countries. In Indonesia,
- BMKG clarified that it is not a heat wave. However, the highest temperatures on record in BMKG (2023) were recorded in Semarang, Surabaya, and Makassar.
- Impacts: heat strain, heat stress, psychosocial disorders, and cognitive performance disruption.
- Hot weather occurs throughout cities, but its effects may differ based on socio-economic status.



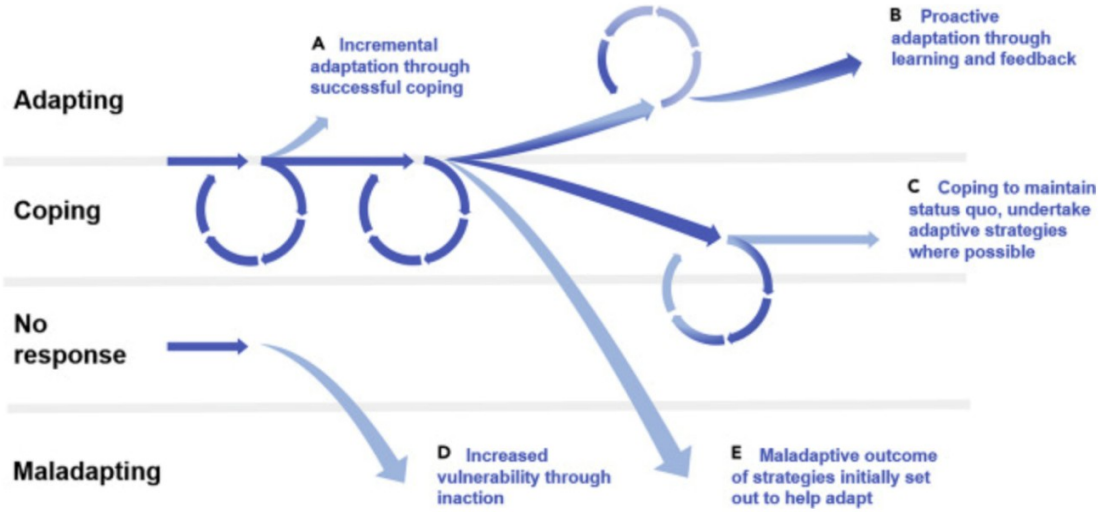




Use of Air Conditioners (AC): Adaptation Strategies

As a climate adaptation strategy

- Life-saving tools from heat, especially for vulnerable groups
- Maintaining quality of life and work productivity



Increasing GHGs Emissions

- High energy users, - increase GHG
- Release of refrigerants (HFC) into the air --> increase GHG

Potentials for Maladaptation

The need for Sustainable Cooling

CO2e emissions from the cooling system

Direct Emission

- CFC, HCFC, and HFC refrigerants have high Global Warming Potential (GWP) values.
- Refrigerant leakage from cooling units.
- Unsustainable management of refrigerants during cooling machine servicing.

Indirect Emission

- Originating from energy consumption in cooling systems.
- In hotels, office buildings, shopping centers: energy consumption for cooling systems can reach 50-60% of the building's total energy consumption.

Principles of Sustainable Cooling

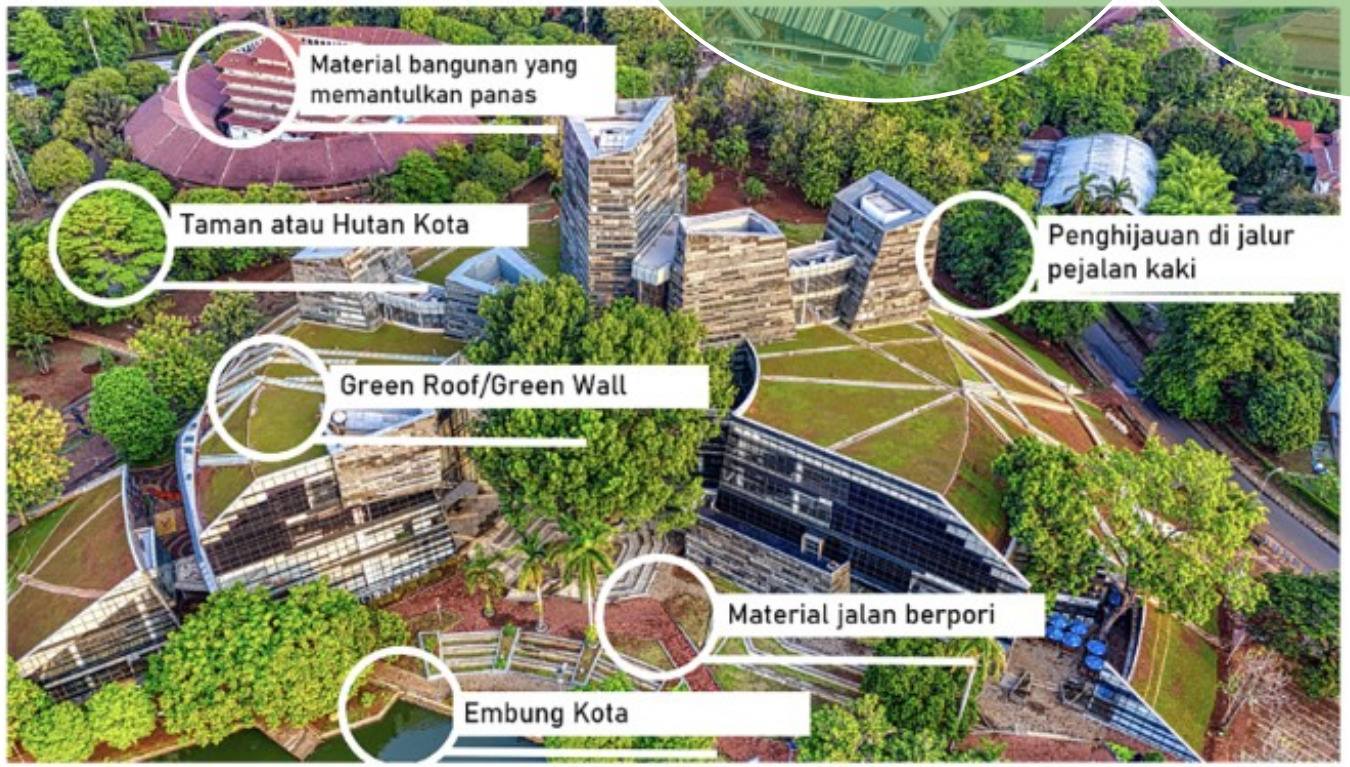
- Energy savings
  - Use energy-efficient ACs, refrigerators, freezers, cold storage, and chillers
  - Reduce thermal load with passive design (ventilation, insulation, shading)
- Environmentally friendly refrigerants
  - Do not damage the ozone layer --> Zero ODP (Ozone Depleting Potential)
  - Do not contribute to global warming --> Low GWP (Global Warming Potential)
- Integration of renewable energy
  - Solar-powered cooling system
- Fair and affordable access
  - Cold storage for traditional fishermen and village cooperatives in need
- Circular economy & end-of-life management
  - Management of used refrigerants (not released directly into the air)
  - Recycling of cooling units

MITIGATION EFFORT

City level (blue/green infrastructure, disaster preparedness)

Community level (vernacular gathering space for shelter)

Individuals (hydration, bathing, air conditioning, fan, clothing adjustments, etc)



Gambar 3.2 *Green* dan *Blue Infrastructure* di Perkotaan (foto oleh Tom Fisk, Pex



Gambar 4.8 Naungan vernakular di Galur, Jakarta

6.4 What can we do?

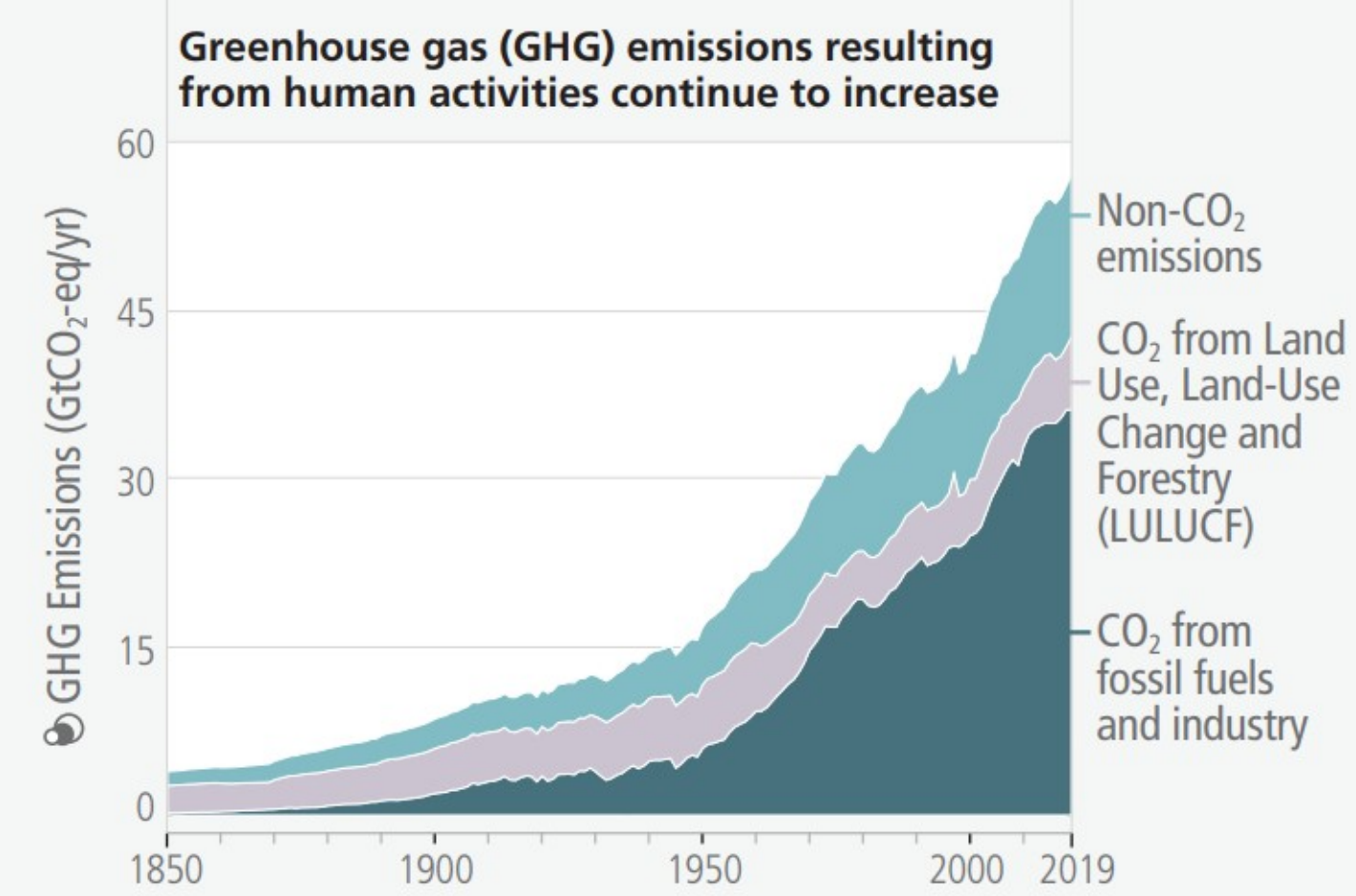
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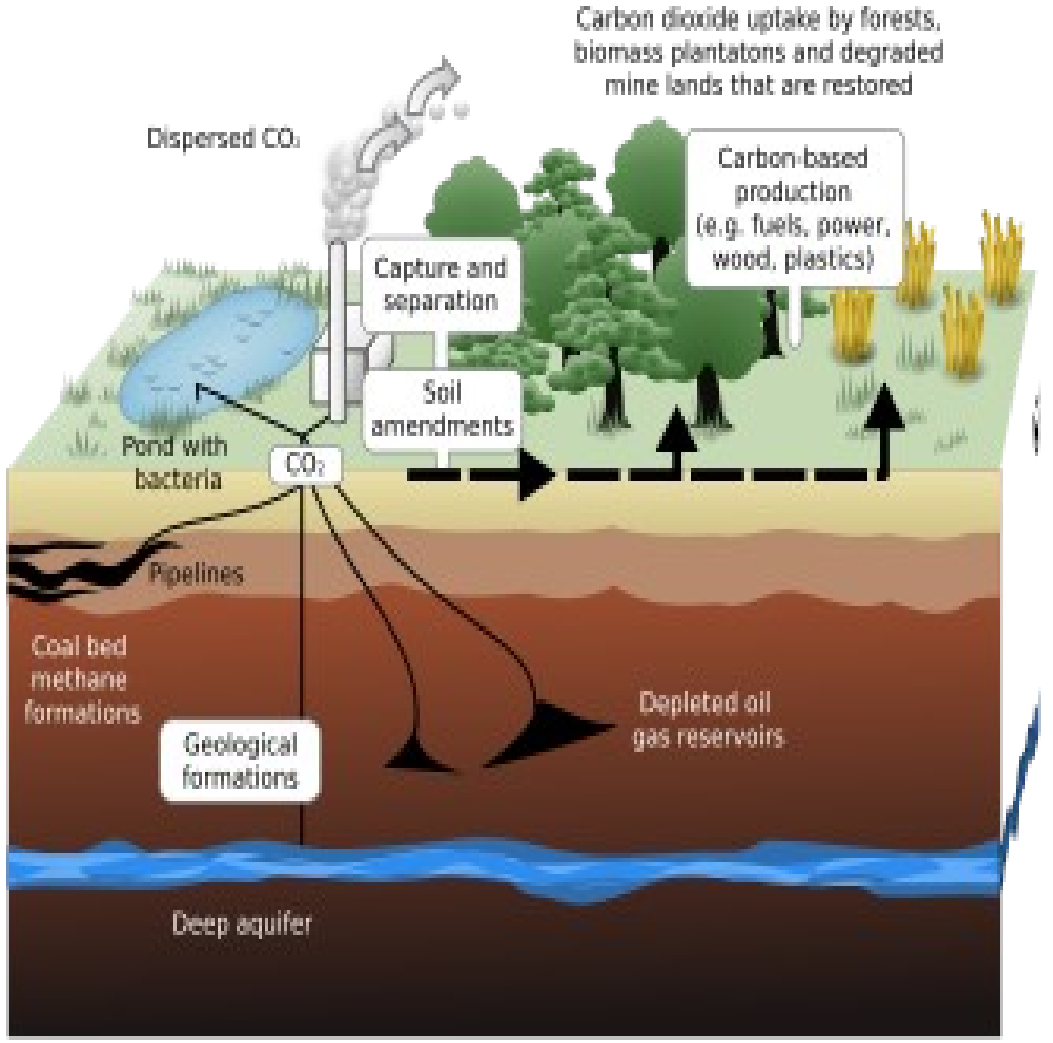
GHGs emissions from human activities

- Global greenhouse gas emissions based on anthropogenic activities
- In 2019, global GHG emissions from:
  - **Energy 34%** (20 $GtCO_2$-eq)
  - **Industrial 24%** (14 $GtCO_2$-eq)
  - **AFOLU (Agriculture, Forestry and Other Land Use) 22%** (13 $GtCO_2$-eq)
  - **Transportation15%** (8,7 $GtCO_2$-eq) ,
  - **Building 6%** (3,3 $GtCO_2$-eq)

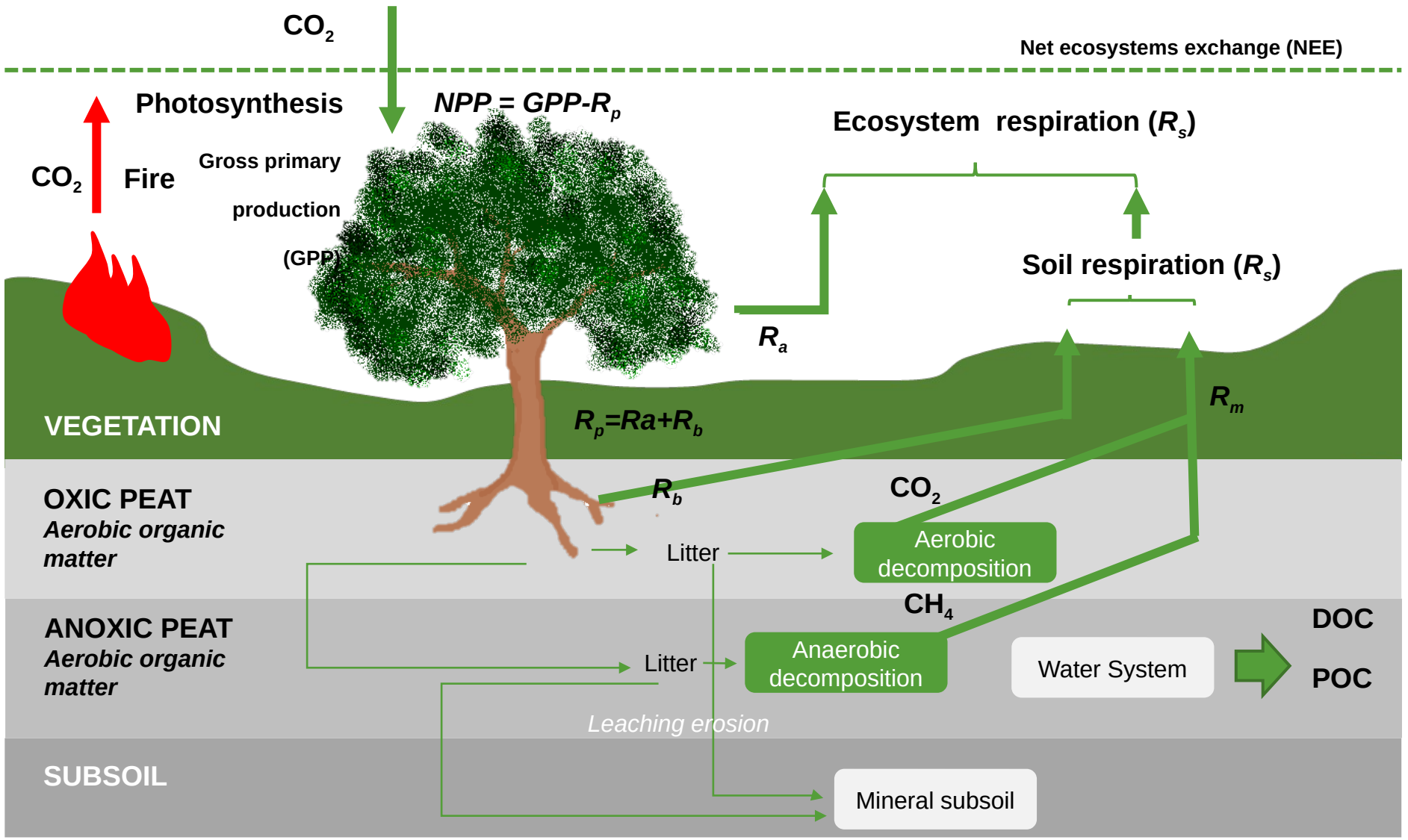


Too much of GHGs, how to reduce the impact? Carbon Sequestration

- “"Carbon sequestration" describes the natural or artificial processes by which CO_2 is removed from the atmosphere or diverted from emission sources, then stored in the ocean, terrestrial environments (vegetation, soil, and sediments), and geological formations.
- Types of carbon sequestration:
  - Biological sequestration
  - Geological sequestration
  - Technological sequestration



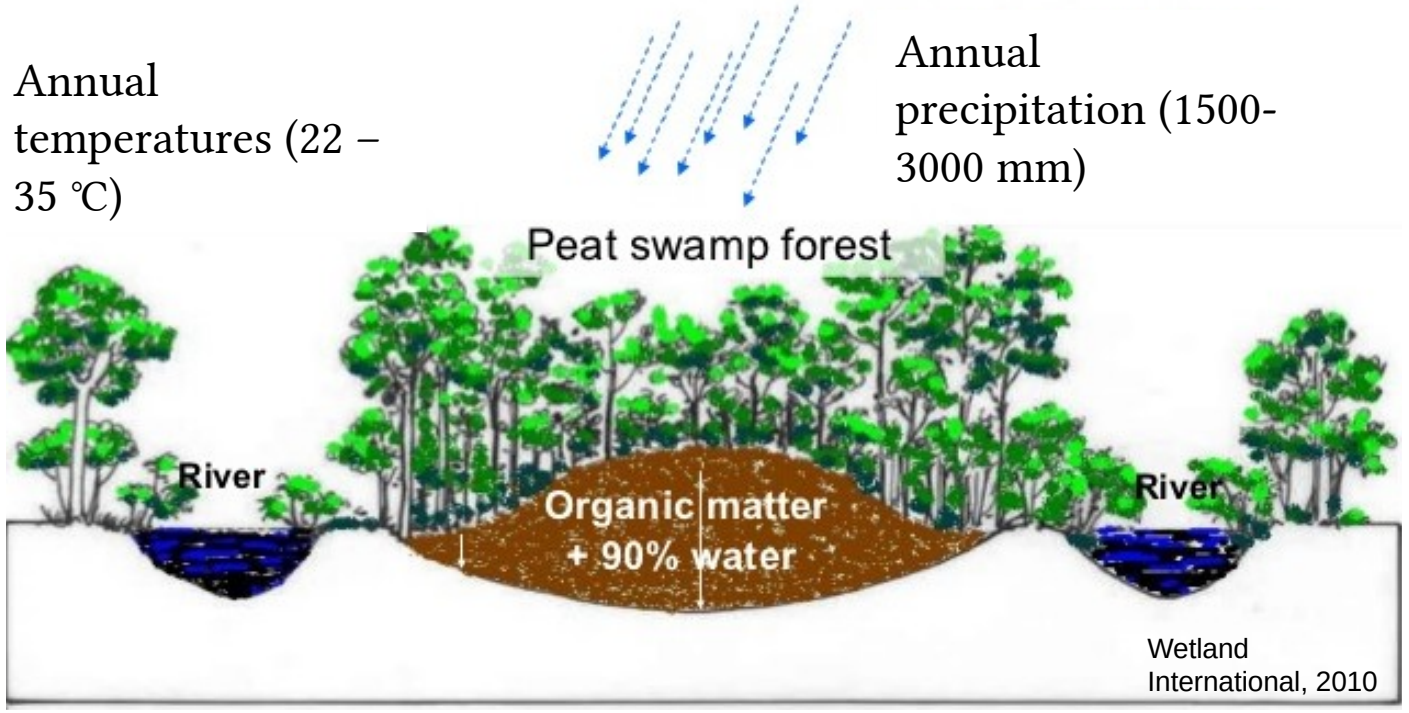
Carbon cycle in Tropical Peatland



(Modify from Luo and Zhuo, 2006; Page et al., 2011)

Tropical peatland formation and carbon capture

Water supply from precipitation and lateral inflows > evapotranspiration--> **water logged condition**, **low rate of decomposition**



Plant production rate > decomposition rate → **accumulation of thick organic soil layers (peat formation)**

Accumulation has started since the late of **Pleistocence** (30 000- 22 000 ^{14}C yr BP), accumulation rate of **7.4 to over 90 g C/m²y** (Page et al., 2004) , deposit **20 m deep** with average of **5.5-7 m→ large carbon store**

Tropical peatlands disturbance in Indonesia

Most rapid tropical peatlands degradation arises in Indonesia (Page et al., 2011)

Deforestation



26.63Mha (47.72%) of the 55.81Mha of Borneo forests recorded in 1973 had been logged over by 2010 (Gaveau et al., 2014)

Drainage



Aerobic condition

Soil decomposition

Soil respiration

Steady emission

Fire



Dry seasons

Peatlands Fire

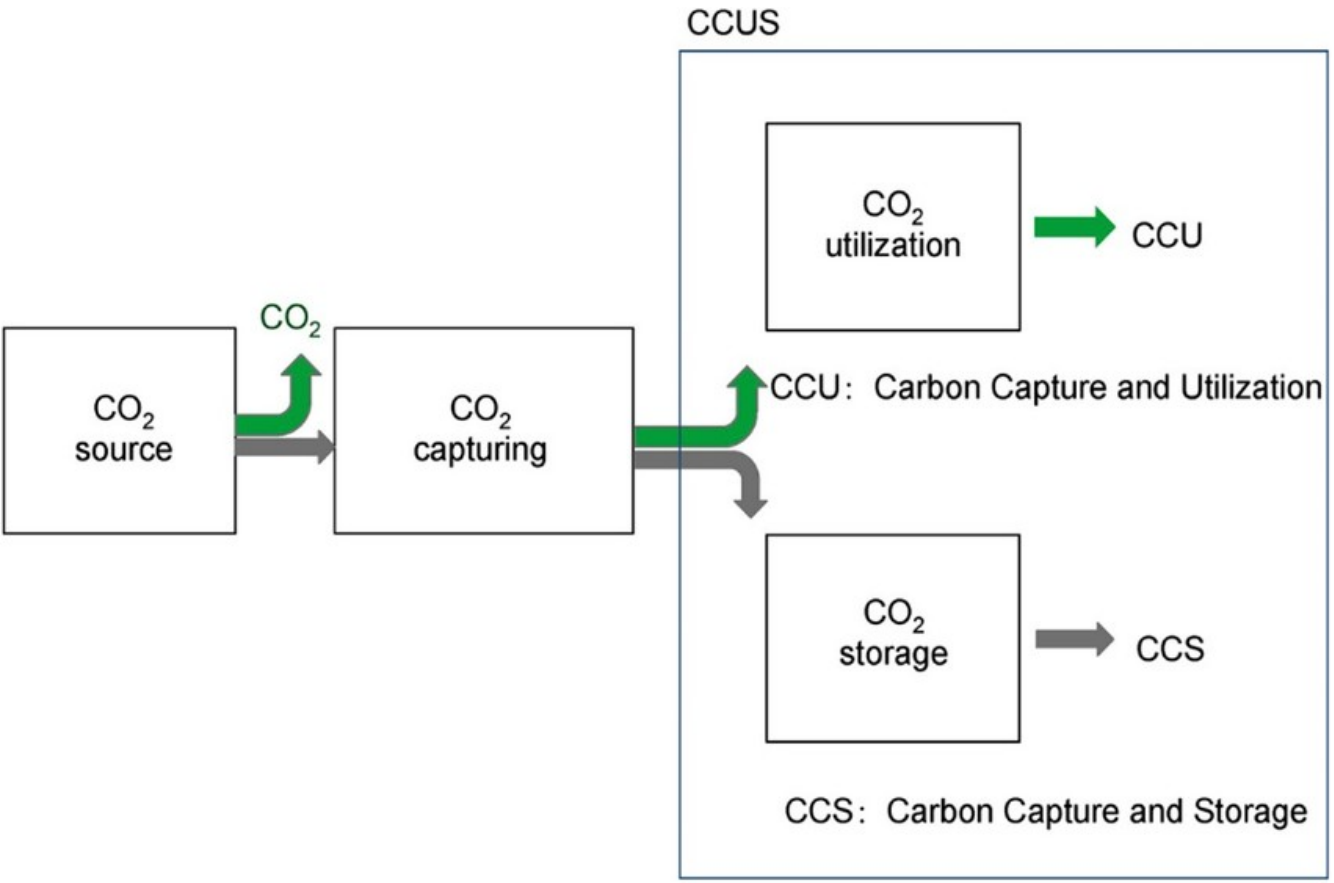
Fire

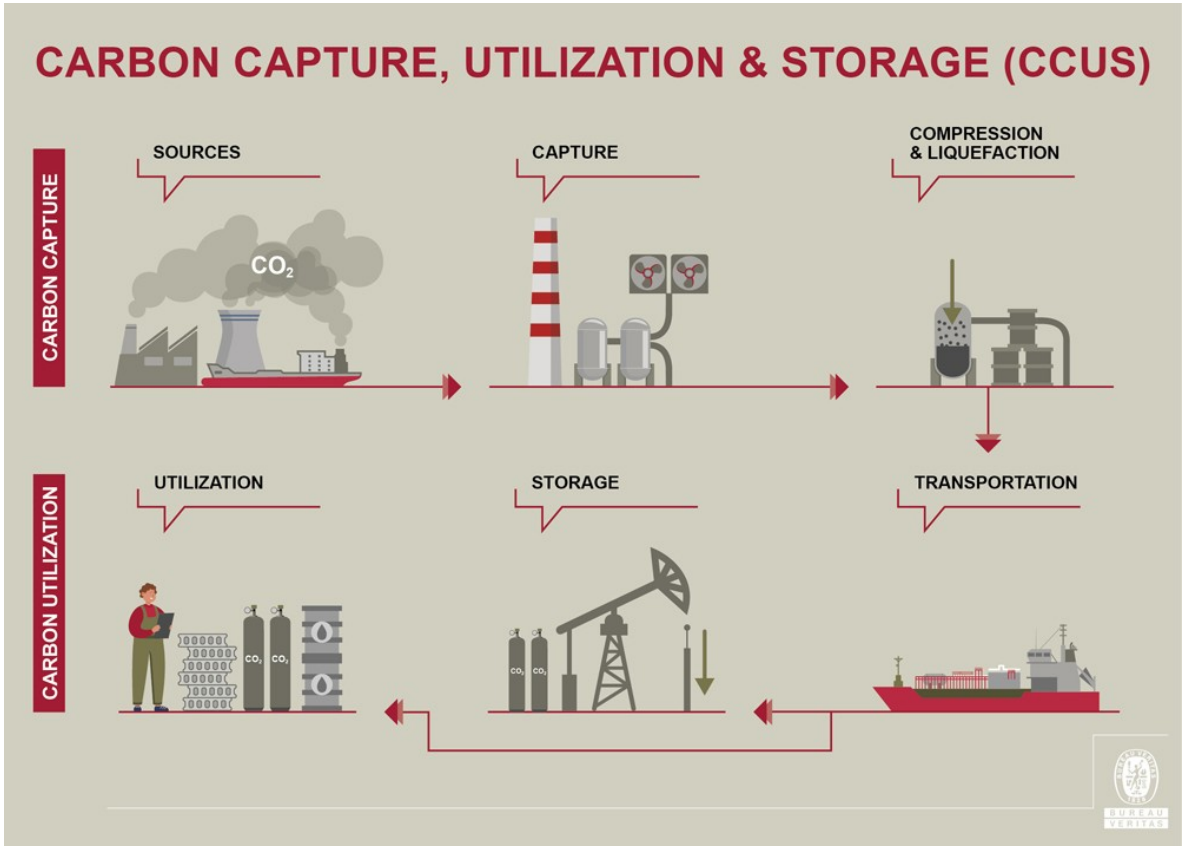
Occasionally emission
Extensive amount

Release of CO_2 to the atmosphere

Too much of GHGs, how to reduce the impact?

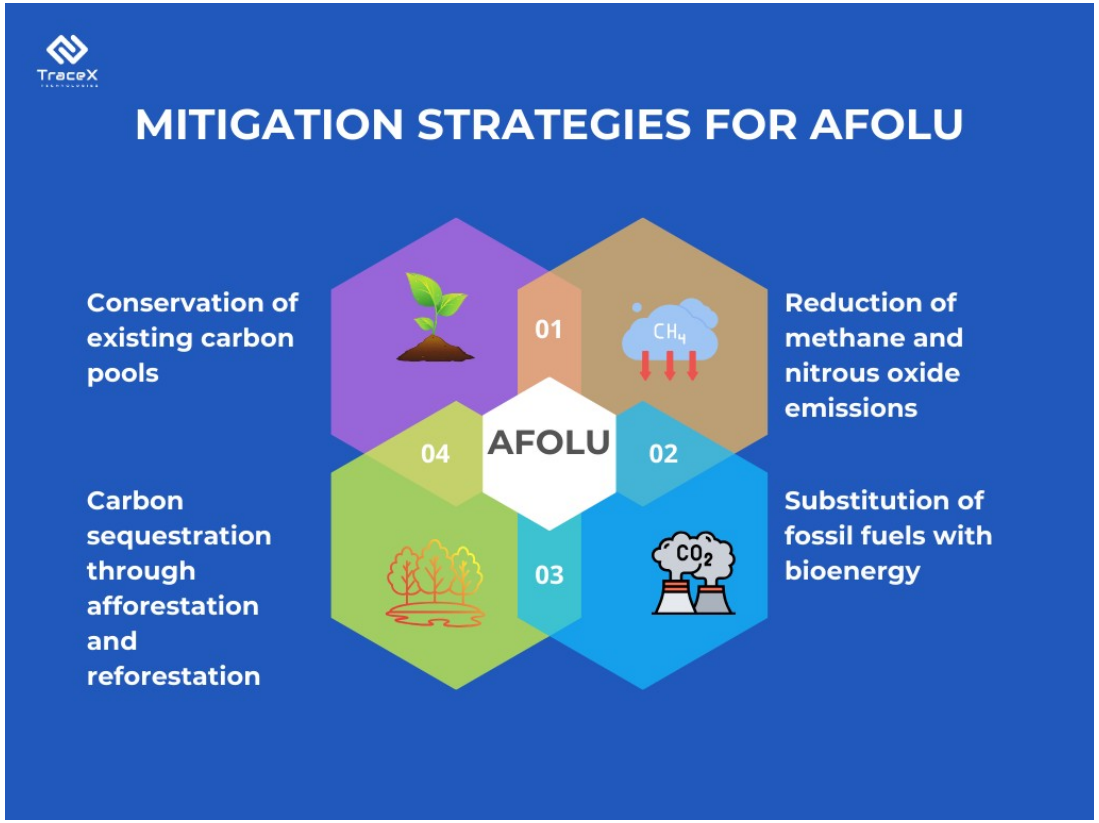
Carbon Capture Utilization & Storage(CCUS)





Reducing GHGs impacts from AFOLU: Agriculture, Forestry, and Other Land Use

- GHG mitigation options in AFOLU, when implemented sustainably, can provide large-scale GHG emission reductions and increase CO_2 sequestration.
- Some strategies with high potential to reduce GHG emissions from the AFOLU sector include:
  - Afforestation, Reforestation, and Forest Ecosystem Restoration
  - Agroforestry practices for food production.
  - Soil carbon management.
  - Use of bioenergy in agricultural practices.



Peatland Fires in Indonesia

National 2 years ago

Residents forced to wear masks as thick haze strikes Central Kalimantan

National 2 years ago

Authorities warned of fires as hot spots detected in Jambi

The Warsi Indonesian Conservation Community warns that at least 38 hot spots have been detected in J...

Indonesia fires forces schools and airport to close in Sumatra

Worsening fires in Indonesia's Sumatra island threaten to cause more diplomatic frictions with neighbouring Malaysia.

Indonesian forests are burning, and Malaysia and Singapore are choking on the fumes

By Jessie Yeung, CNN

Updated 0130 GMT (0930 HKT) September 12, 2019

Singapore shuts schools as pollution from Indonesian fires intensifies

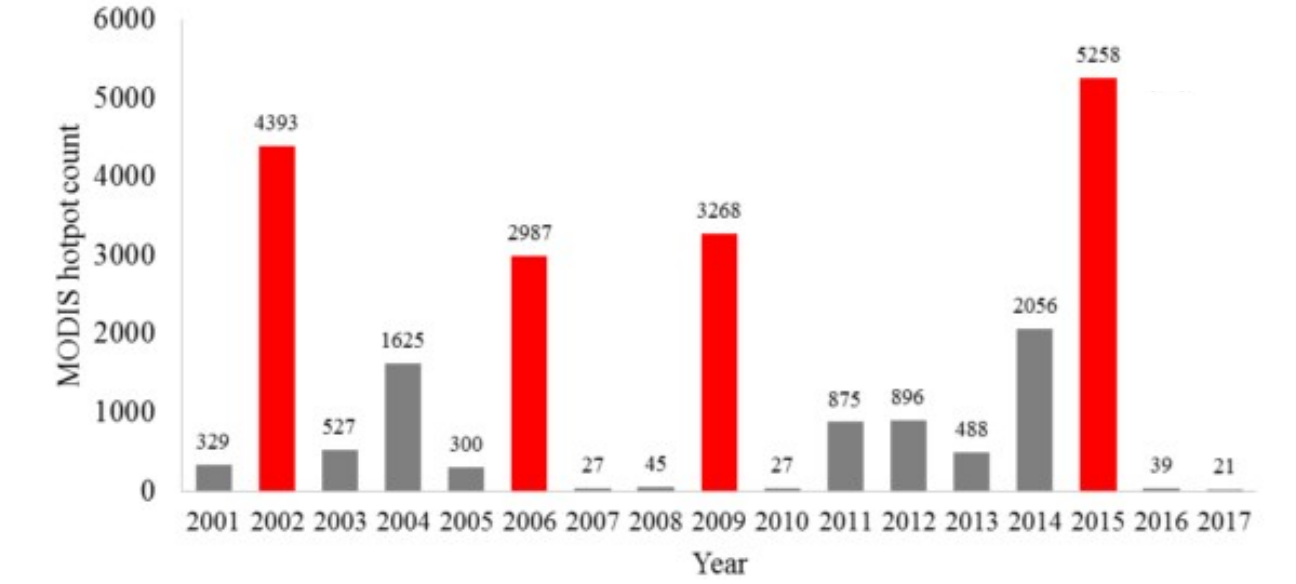
By Georgia McCafferty, for CNN

Updated 1012 GMT (1812 HKT) September 25, 2015

NEWS • NATIONAL

Smoke thickens, claims another life

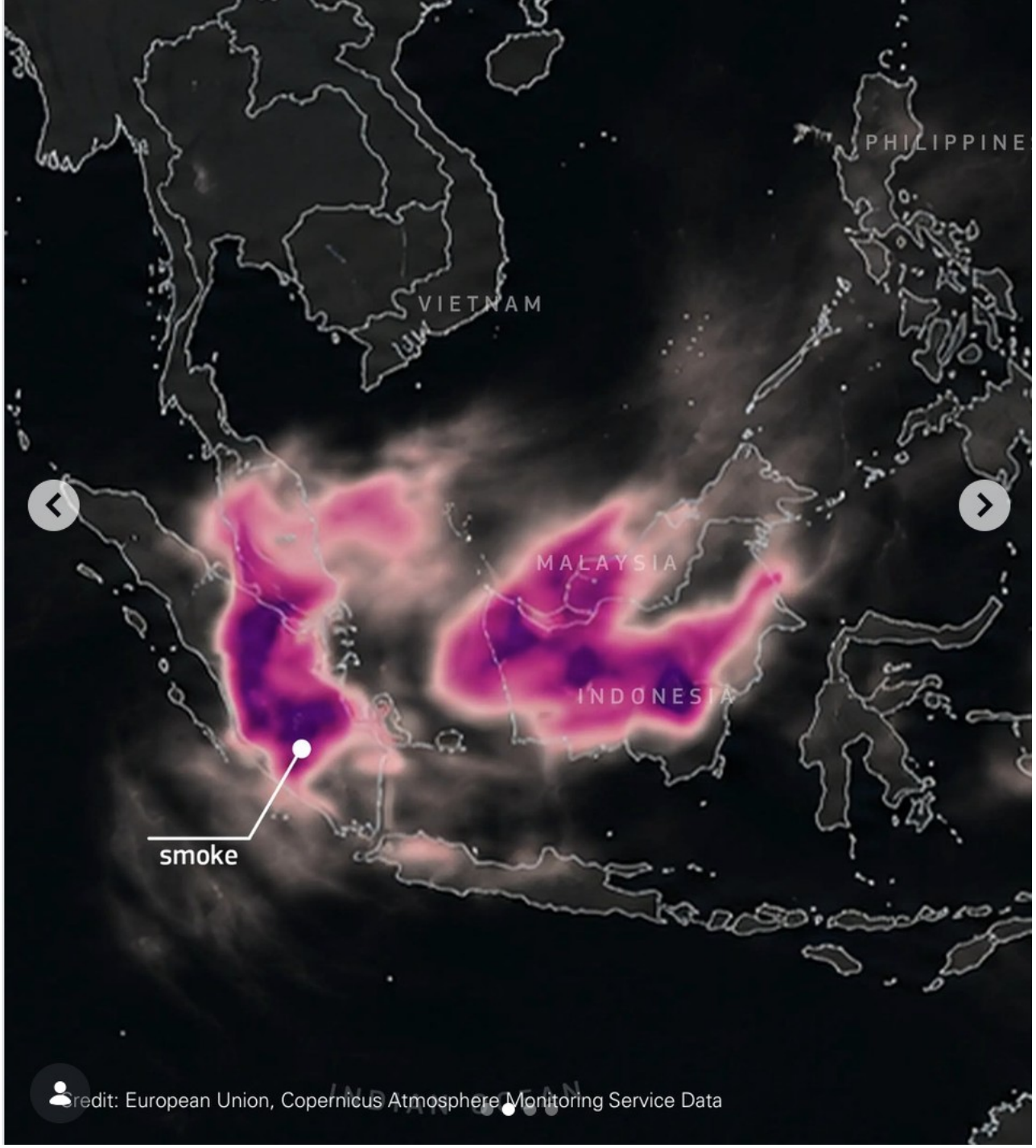
Types of land Uses that lead Peatland Fires in Indonesia



MODIS hotspots counts in a radius distance of 50 km from Palangka Raya, Central Kalimantan over 2001 to 2017 (Red color indicating ENSO year)

Peatland fires occurs every year especially during the dry season, when the temperature is high, and precipitation number is low.

Other Land uses that lead to Disaster: Wildfire in Indonesia



Discussion:

- Do you feel that climate change has brought changes to your life behavior?
- Do you recognize behaviors or habits that are climate-friendly?
- What will you do to support a climate-friendly lifestyle?

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